# AQA (Trilogy) Combined Science

# GCSE Unit 6.2 Electricity - **Answers**

## Test (Levels 4 – 9)

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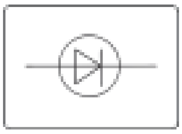
|---|---|---|
| **1.** | | **(4 marks total)** |
| a. | **lamp**<br>**fuse**<br>**LDR thermistor**<br>**diode**<br>one mark per correct answer **maximum 3** | **(3 mark)** |
| b. | Delocalised electrons | **(1 mark)** |
| **2.** | | **(7 marks total)** |
| a. | Total power = 500+1500+1600+2500+700 = 6800 watts<br>Current = 6800 ÷ 230<br>= 29.57 A (2dp) - **(accept 29.565)**<br>This is less than 30A so it is safe to use<br>**Maximum 1 mark for an unqualified yes**<br>**Maximum 2 marks for 29.57 with no working or qualifying yes** | **(1 mark)**<br>**(1 mark)**<br>**(1 mark)**<br>**(1 mark)** |
| b. | **Q=Ixt** so, 2400 = I x 150<br>I = 2400/150 = 16<br>unit = amps<br>**Allow** 1 mark for correct substitution but no answer | **(1 mark)**<br>**(1 mark)**<br>**(1 mark)** |
| **3.** | | **(6 marks total)** |
| a. | **V=IR or** potential difference = current x resistance | **(1 mark)** |
| b. | 3.6 = 0.17 x R<br>R = 3.6 ÷ 0.17<br>R = 21.18 (2dp) - **allow** 21 or 21.2<br>**Allow** answer showing no working out for 3 marks | **(1 mark)**<br>**(1 mark)**<br>**(1 mark)** |

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| c. | As resistance increases, current decreases...<br>...this makes the bulb dimmer<br>**Accept** opposite argument | **(1 mark)**<br>**(1 mark)** |



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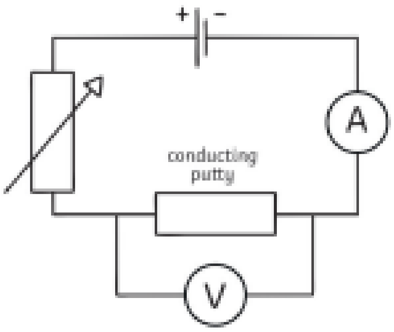
| **4.** | | **(9 marks total)** |
| a. |  | **(2 marks)** |
| b. | A variable resistor can be added<br>**Accept** change resistance, or<br>Change number of cells/batteries<br>**Accept** alter voltage/potential difference | **(1 mark)**<br><br>**(1 mark)** |
| c. | 0.24 (Ω) | **(1 mark)** |
| d. | V=IR<br>so V = 0.15 x 0.24 = 0.036 (v)<br>**Allow** 1 mark for correct substitution without answer **Allow** 2 marks if correct calculation with the incorrect answer from c. ecf | **(1 mark)**<br>**(1 mark)** |
| e. | As the putty gets thicker, the resistance increases (or reverse) | **(1 mark)** |
| f. | Any one from:<br>Inaccurate measurements of current/length/voltage<br>Putty not the same thickness throughout the cylinder **Accept sensible qualified systematic or human errors** | **(1 mark)** |
| **5.** | | **(4 marks total)** |
| a. | The voltage across the cables is increased<br>This decreases the current (through the cables)<br>To reduce heat energy loss in the wires | **(1 mark)**<br>**(1 mark)**<br>**(1 mark)** |
| b. | Reduces the voltage to allow safe domestic use | **(1 mark)** |

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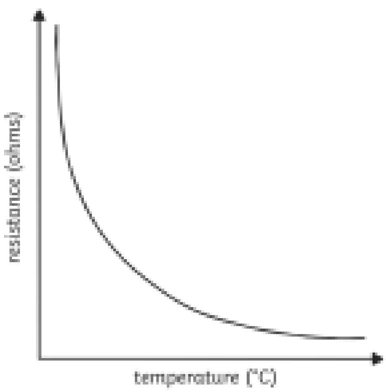
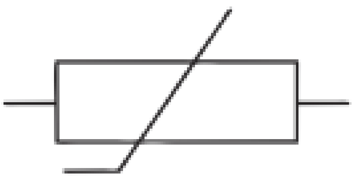
| **6.** | | **(7 marks total)** |
| a. | Series circuits have components in a loop, parallel circuits are branched<br>**Allow** brighter bulbs than a series circuit | **(1 mark)** |
| b. | Parallel circuits allow independent switching<br>**Allow**: brighter bulbs than in a series circuit or if one bulb blows the rest stay on | **(1 mark)** |
| c. | Fuse | **(1 mark)** |
| d. | **A** - same as/less than | **(1 mark)** |
| e. | Current is continuously...<br>...changing direction | **(1 mark)**<br>**(1 mark)** |
| f. | 50Hz | **(1 mark)** |
| **7.** | | **(3 mark total)** |
| a. | A circuit which automatically turns a heating system on and off | **(1 mark)** |
| b. |  | **(1 mark)** |
| c. |  | **(1 mark)** |
| **8.** | | **(10 marks total)** |

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| a. | Component C is a variable resistor and will allow a range of Potential difference values to be obtained<br>**Accept** will increase/decrease the current/voltage/pd/resistance **Accept** control or change<br>**Both parts needed for full mark** | **(1 mark)** |
| b. | The bulb gets hotter/temperature of the bulb increases | **(1 mark)** |
| c. | Power = pd x current (or P=VI)<br>So, 9 x 2.5 = 22.5<br>Watts (W) | **(1 mark)**<br>**(1 mark)** |
| d. | **QWC**<br>5-6 marks shows a detailed comparison of both cost and energy efficiency<br><br>3-4 marks shows a clear comparison of cost OR energy efficiency<br><br>1-2 marks shows a basic comparison of an aspect of either cost OR energy efficiency<br><br>Examples<br><br>Energy efficiency<br><br>LED has same output but less power<br><br>LED is 22% more efficient<br><br>LES has less wasted energy<br><br>LED uses a smaller current<br><br>LED produces less heat<br><br>Cost<br><br>LED is approximately 6 times more expensive<br><br>LED cheaper to run than a halogen<br><br>18 halogens needed to last the same amount of time as one<br><br>LED 18 halogens cost £36,90 | **(6 marks)** |